

Saving money for consumers in delivering high-quality streaming video

A 3-way graceful tunable tradeoff exploiting:

- Video compressibility
- Consumer usage pattern
- Monthly cellular data quota

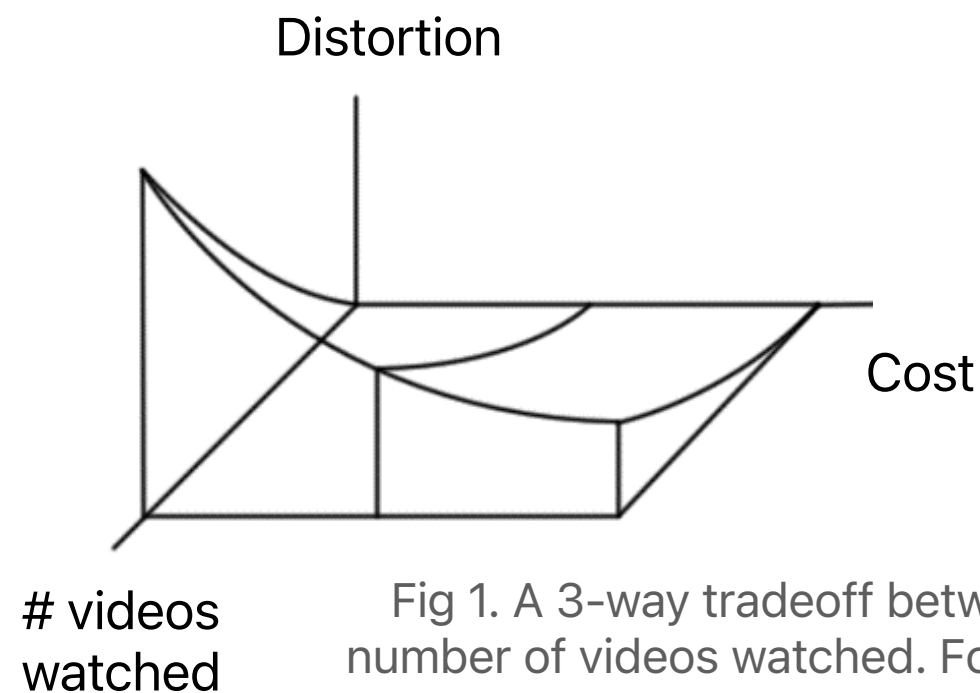


Fig 1. A 3-way tradeoff between distortion, cost, and the number of videos watched. For a fixed cost, QAVA enabled a user to watch the desired videos while minimising distortion.

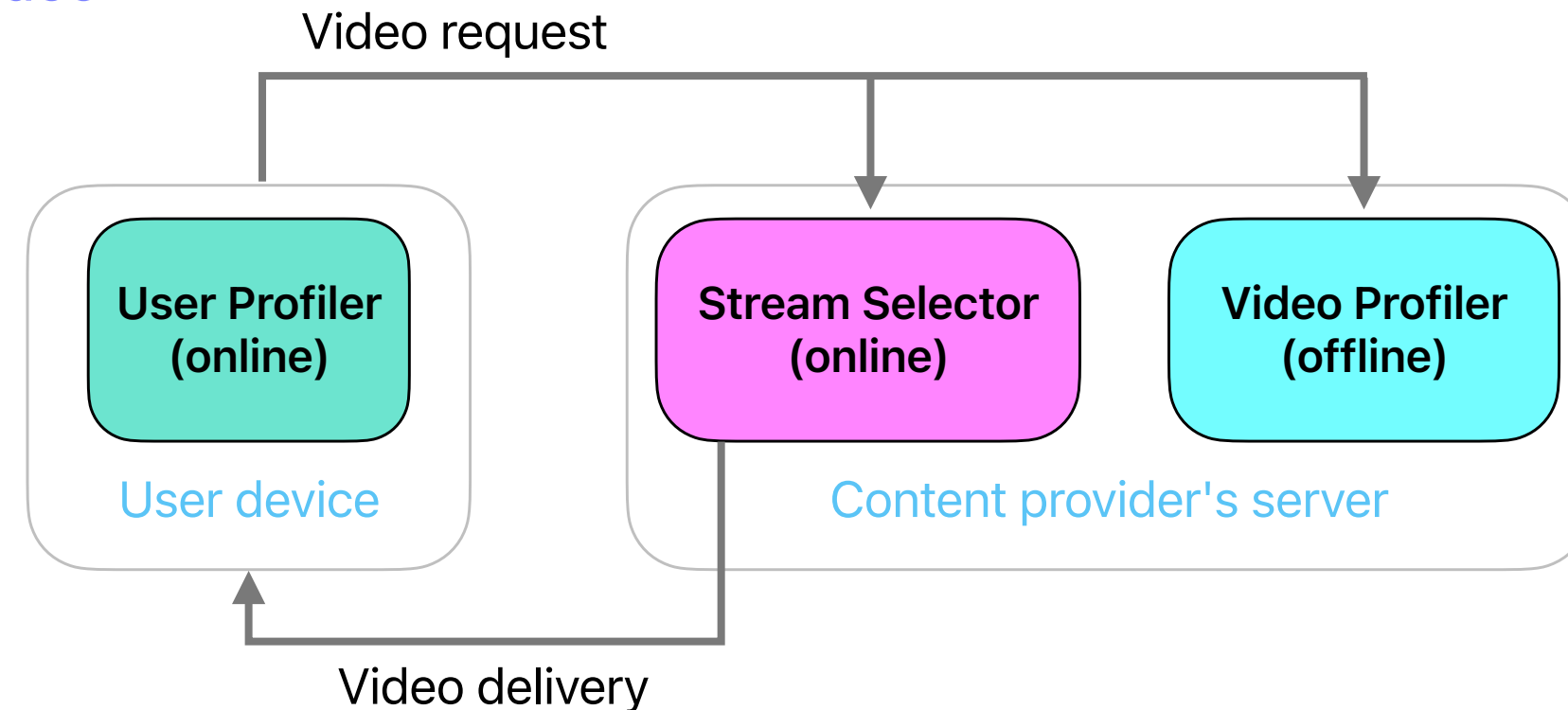


Fig 2. QAVA's modular system architecture: The user profiler sits on a user device, whereas the stream selector and video profiler are on a content provider's server. A video request originating from a user travels through the access network and the backbone to the server, which then runs a stream selection algorithm online to choose an optimal bit rate for the user.